

Complex Networks - Project

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1 In short

1. Identify a question for which a network perspective can be insightful. Ideally, you should favour open, unsettled questions, possibly inspired by the news (some ideas are given below, but they are not limitative).
2. Create a network related to this question and analyse it in light of what you learned in the lectures.
3. Write a report summarising your findings¹ with respect to the question identified in step 1, and prepare an oral presentation.

2 Guidelines (*optional*)

2.1 Creating your own network

Once you have identified the question you want to address, collect relevant data to create a network related to this question.

For this project, you need to create your *own* network(s), by

- collecting data external sources, either obtained online (e.g.,² using an API) or by contacting someone who can provide you with data that are not openly available
- starting from an existing dataset which is not in graph format (e.g., data of taxi trips including source and destination to create a transport network, dataset of users / music to build a network of music similarity, *MovieLens* dataset³ for movies, adjacent amino-acids in proteins, etc.). Extract a list of edges from this dataset and create the network using e.g. `networkx`. Feel free to ask for help and advice.
- simplifying or enriching an existing network. (For geographic data, SIG tools such as QGIS may be helpful.)

¹There is *no need to fully settle the question*. We only request that you honestly expose your progress towards the solution, and how the network perspective can shed light on it.

²Prompting an LLM is acceptable, provided you specify what you did and you verify the consistency of the output.

³<https://grouplens.org/datasets/movielens/latest/>

Your report will start by introducing the network you created: What do the nodes/edges represent? Comment on the choices or assumptions you made to create the network and their potential impact.

2.2 Format of the report

You can choose either to send a Jupyter Notebook or a PDF file. If a Notebook, it should be easy to read and run (functions in external files can be called; the data must be attached). If you send a PDF, join the code that you wrote and used in another document

2.3 Visualization

Remember that a nice graph is worth 1000 words. If you want to include graphs drawn with Gephi, you can include them in the notebook, but do not forget to include the original image when you send your report. You can also learn to use the pyvis library⁴

2.4 What question should I address? And how?

There is no strong constraint on the question that you will address. We simply request that (i) it *might* be worth addressing it from a network perspective (if it turns out the network analysis actually does not help, this is okay, as long as you explain your endeavours and how you arrived at this conclusion), (ii) the constructed network and/or its analysis are not already widely available.

3 Indicative (and non-exhaustive) list of possible questions

3.1 Pipeline network

On September 22, 2026, gas leaks appeared on the Nord Stream 1 and 2 pipelines, hinting at a possible sabotage. *What countries were directly/indirectly impacted? How central in the (past/present/hypothetical future) pipeline network⁵ are the pipelines segments that suffered from these leaks?*

3.2 Submarine communications cables

Submarine cables⁶ are the backbone of the intercontinental telecommunication network, but they are regularly damaged by human activity, unintentionally or through malicious action. *How vulnerable is this network to the failure of*

⁴<https://pyvis.readthedocs.io/en/latest/>

<https://memgraph.com/blog/gas-pipelines-in-europe>

⁵<https://mondediplo.com/maps/gas-pipelines>

⁶<https://www.submarinecablemap.com/>

individual cables? Which cables most deserve to be protected? Can the location of cable faults tell us if, statistically, malicious action is likely to be involved?

3.3 Great Fear

Zapperi et al.⁷ (*Nature*, 2025) studied the propagation of the 1789 Great Fear in France, using an epidemiological model. The Python scripts they used are available as supplemental material. *How can one characterise the network of locations where the Great Fear spread? Are there locations which one can presume played a major role / little role in the propagation of this Great Fear?*

3.4 Sea lanes / Marine traffic routes

In the aftermath of the Israel-American strikes in Iran in 2026, the Hormuz strait has been partly closed to merchant ships and tankers, which has affected the global economy, to variable extents depending on the country. Construct a network of countries (or world regions) in which edges represent major sea lanes⁸ (possibly for one specific type of staples). *Does this network help understand the economic impact of the partial closure of the Strait of Hormuz?*

3.5 Forest fires

The French territory was severely hit by megafires in summer 2026.

Land cover maps⁹ enable one to create a graph of land patches connected by flammable areas (e.g., pine forests). *If the weather conditions are unknown, what properties of this network control the risks that fire may reach a given patch (without human intervention)? To what extent does this highly simplified approach reflect the propagation of forest fires¹⁰ in France in summer 2026?*

3.6 Power grid

On April 28, 2025, a power blackout occurred on the Spanish and Portuguese power grids. *Can the electricity transmission grid¹¹ help understand the spatial extent of the blackout? How can one quantify how interlocked the Spanish and Portuguese power grids are, as compared to the Spanish and French ones?*

⁷<https://www.nature.com/articles/s41586-025-09392-2>

⁸<https://www.marinetraffic.com/en/ais/home/centerx:53.1/centery:25.9/zoom:10>
<https://resourcewatch.org/data/explore/com012-Global-Shipping-Lanes>

⁹<https://www.data.gouv.fr/datasets/corine-land-cover-occupation-des-sols-en-france>

¹⁰<https://feuxdeforet.fr/cartes/historique/>

https://www.lemonde.fr/les-decodeurs/article/2026/07/23/la-carte-en-temps-reel-des-incendies-actifs-et-des-surfaces-brulees-en-france-et-en-europe_6723063_4355771.html?srsltid=AfmBOoqq_QYJhFsQzO5msWo5krB06K6LIXVsE0u8rcuWqqNxrz3enGFq

¹¹ See e.g. <https://www.entsoe.eu/data/map/>

3.7 Adjacent street names

Most streets have names. Identify the most common street names in France. Construct a network of street names in which edges connect adjacent streets for several large French cities and comment on their (dis)similarity.